



SEAGLASS FOUNDRY
Forged For Speed. Refined by Craft.

ForgeGIS™

GPU-Accelerated Geospatial Compute for the Agent Era

To our knowledge, the only GPU-accelerated geospatial MCP server in existence. 220 operations spanning the working surface of professional GIS analysis. Pure Java. Zero native dependencies. Designed from the start for both programmatic pipelines and AI-agent-driven workflows.

220

**user-facing
operations**

across 11 categories

218 / 220

**accessible to
AI agents via MCP**

*62 MCP tools · 133-op pipeline
DSL*

14×-40×

**multi-stage pipeline
speedup vs. GDAL CLI**

*May 2026 validation run, 4096²
rasters*

What it is

ForgeGIS is the GPU-accelerated compute substrate for geospatial analysis, written entirely in Java with no JNI and no native dependencies. It exposes 220 operations across terrain, bathymetry, hydrology, RF/visibility, spectral and ML, spatial geometry, filtering, raster I/O, routing, point cloud, and temporal analysis — callable directly from Java for programmatic pipelines, and exposed to AI agents through a dual-surface MCP server.

Why it matters

Today's geospatial stack was built for a human analyst at a desktop. Agents need different primitives — composable, callable, fast enough that a multi-step pipeline doesn't time out. ForgeGIS provides those primitives without forcing organizations to choose between agentic workflows and the deterministic, scripted control their environments require.

How it's built differently

Data stays GPU-resident across operations rather than round-tripping through host memory or disk between stages. A single Java jar drops into existing Maven builds — no GLIBC pinning, no shared-library vetting, no native binary distribution to coordinate in air-gapped or regulated environments.

WHO FORGEGIS IS FOR

Defense, intelligence, and aerospace contractors

Pure-Java deployment without native dependencies means ForgeGIS drops into existing JVM environments — including air-gapped and classified deployments — without coordinating GLIBC compatibility or shared-library signing. Numerical outputs are verified deterministic across surfaces, preserving audit and reproducibility guarantees when operations are invoked by AI agents.

Builders of AI agents for geospatial workflows

The dual-surface MCP server exposes a pipeline DSL that composes 133 operations into multi-step workflows agents construct dynamically, plus 54 curated intent tools for typed single-call analytical tasks. Agents reach professional-grade GIS analysis through MCP without needing to know geospatial internals.

Mid-market geospatial SaaS, climate, and agtech platforms

GPU-accelerated geospatial compute at server-class performance on commodity laptop hardware — the entire May 2026 validation run was measured on a single laptop with an RTX 5070. Drop ForgeGIS into existing Java services without replatforming on CUDA or coordinating native binary distribution.

PERFORMANCE HIGHLIGHTS

6.88×

geometric mean across 143 catalog
operations vs. GDAL CLI
(127 wins of 143; format-conversion round-trips
excluded)

14.46×
39.75×

multi-stage GPU-resident pipelines at
2048²–4096² rasters
(TerrainPipeline, WarpPipeline,
AlgebraPipeline)

13.76×
36.94×

per-op median speedups across five real-
world DEM datasets
(Copernicus and ETOPO, 2048²–4096×4096
cells)

WHAT'S IN THE CATALOG

Terrain · Bathymetry & Ocean · Hydrology · Visibility & RF · Spectral & ML ·
Spatial Geometry · Filtering, Focal, & Morphology · Raster Management, Vectorize, & I/O ·
Routing · Point Cloud · Temporal

220 operations total. 218 accessible via the MCP server (62 tools — 54 intent tools, 5 primitive handle and session tools, 3 pipeline-escape tools). The complete enumerated catalog and per-operation validation oracles are documented in the full Technical Brief.

Get the full Technical Brief

A 21-page technical brief with the complete 220-operation catalog, validation methodology, per-category benchmark breakdowns, and architectural deep-dive is available on request.

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